IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS

COMMUNICATIONS

Monday, 28 April 2025, 2:30 PM

Time allowed: 2:00 hours

There are THREE questions on this paper.

Answer ALL questions.

Any special instructions for invigilators and information for candidates are on page 1.

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EXAM QUESTIONS

Information for Students

Fourier Transform Pairs

Pair Number	$x(t)$	$X(f)$
1.	$\Pi\left(\frac{t}{\tau}\right)$	$\tau \operatorname{sinc} \tau f$
2.	$2W \operatorname{sinc} 2Wt$	$\Pi\left(\frac{f}{2W}\right)$
3.	$\Lambda\left(\frac{t}{\tau}\right)$	$\tau \operatorname{sinc}^2 \tau f$
4.	$\exp(-\alpha t)u(t),\ \alpha > 0$	$\frac{1}{\alpha + j2\pi f}$
5.	$t \exp(-\alpha t)u(t),\ \alpha > 0$	$\frac{1}{(\alpha + j2\pi f)^2}$
6.	$\exp(-\alpha\|t\|),\ \alpha > 0$	$\frac{2\alpha}{\alpha^2 + (2\pi f)^2}$
7.	$e^{-\pi(t/\tau)^2}$	$\tau e^{-\pi(f\tau)^2}$
8.	$\delta(t)$	1
9.	1	$\delta(f)$
10.	$\delta(t - t_0)$	$\exp(-j2\pi f t_0)$
11.	$\exp(j2\pi f_0 t)$	$\delta(f - f_0)$
12.	$\cos 2\pi f_0 t$	$\frac{1}{2}\delta(f - f_0) + \frac{1}{2}\delta(f + f_0)$
13.	$\sin 2\pi f_0 t$	$\frac{1}{2j}\delta(f - f_0) - \frac{1}{2j}\delta(f + f_0)$
14.	$u(t)$	$(j2\pi f)^{-1} + \frac{1}{2}\delta(f)$
15.	$\operatorname{sgn} t$	$(j\pi f)^{-1}$
16.	$\frac{1}{\pi t}$	$-j \operatorname{sgn}(f)$
17.	$\hat{x}(t) = \frac{1}{\pi}\int_{-\infty}^{\infty} \frac{x(\lambda)}{t - \lambda}\, d\lambda$	$-j \operatorname{sgn}(f) X(f)$
18.	$\sum_{m=-\infty}^{\infty} \delta(t - mT_s)$	$f_s \sum_{m=-\infty}^{\infty} \delta(f - mf_s),$ $f_s = T_s^{-1}$

Useful Relations and Formulas

$$\tan(x+y) = \frac{\tan x + \tan y}{1 - \tan x \tan y}$$

$$\tan(x-y) = \frac{\tan x - \tan y}{1 + \tan x \tan y}$$

$$\tan(\frac{\pi}{2} - x) = \cot x$$

$$\tan 2x = \frac{2\tan x}{1-\tan^2 x} = \frac{2\cot x}{\cot^2 x - 1} = \frac{2}{\cot x - \tan x}$$

$$\tan\frac{x}{2} = \pm\sqrt{\frac{1-\cos x}{1+\cos x}} = \frac{\sin x}{1+\cos x} = \frac{1-\cos x}{\sin x}$$

$$\sin x \cos y = \frac{1}{2}[\sin(x+y) + \sin(x-y)]$$

$$\cos x \sin y = \frac{1}{2}[\sin(x+y) - \sin(x-y)]$$

$$\sin x \sin y = \frac{1}{2}[\cos(x-y) - \cos(x+y)]$$

$$\cos x \cos y = \frac{1}{2}[\cos(x+y) + \cos(x-y)]$$

$$\sin x + \sin y = 2\sin\frac{x+y}{2}\cos\frac{x-y}{2}$$

$$\sin x - \sin y = 2\cos\frac{x+y}{2}\sin\frac{x-y}{2}$$

$$\cos x + \cos y = 2\cos\frac{x+y}{2}\cos\frac{x-y}{2}$$

$$\cos x - \cos y = -2\sin\frac{x+y}{2}\sin\frac{x-y}{2}$$

Differentiation Rule of Leibnitz. Let $F(z) = \int_{a(z)}^{b(z)} f(x,z)dx$. We have

$$\frac{dF(z)}{dz} = \frac{db(z)}{dz}f(b(z),z) - \frac{da(z)}{dz}f(a(z),z) + \int_{a(z)}^{b(z)} \frac{\partial f(x,z)}{\partial z}dx$$

Entropy. For a random variable X defined over alphabet $\mathscr{X}$, its entropy is defined as

$$H(X) = -\sum_{x\in\mathscr{X}} p(x)\log_2 p(x).$$

Fourier Transform Theorems[a]

Name of Theorem		
1. Superposition (a_1 and a_2 arbitrary constants)	$a_1x_1(t) + a_2x_2(t)$	$a_1X_1(f) + a_2X_2(f)$
2. Time delay	$x(t - t_0)$	$X(f)e^{-j2\pi ft_0}$
3a. Scale change	$x(at)$	$\lvert a\rvert^{-1}X\left(\frac{f}{a}\right)$
b. Time reversal	$x(-t)$	$X(-f) = X*(f)$
4. Duality	$X(t)$	$x(-f)$
5a. Frequency translation	$x(t)e^{j\omega_0 t}$	$X(f - f_0)$
b. Modulation	$x(t)\cos\,\omega_0 t$	$\frac{1}{2}X(f - f_0) + \frac{1}{2}X(f + f_0)$
6. Differentiation	$\frac{d^n x(t)}{dt^n}$	$(j2\pi f)^n X(f)$
7. Integration	$\int_{-\infty}^{t} x(t')\,dt'$	$(j2\pi f)^{-1}X(f) + \frac{1}{2}X(0)\delta(f)$
8. Convolution	$\int_{-\infty}^{\infty} x_1(t - t')x_2(t')\,dt'$	$X_1(f)X_2(f)$
	$= \int_{-\infty}^{\infty} x_1(t')x_2(t - t')\,dt'$	
9. Multiplication	$x_1(t)x_2(t)$	$\int_{-\infty}^{\infty} X_1(f - f')X_2(f')\,df'$
		$= \int_{-\infty}^{\infty} X_1(f')X_2(f - f')\,df'$

Multivariate Gaussian density. Let $\boldsymbol{X} \in \mathbb{R}^d$ be a d-variate Gaussian random variable with mean vector $\boldsymbol{\mu}$ and covariance matrix $\boldsymbol{\Sigma}$, denoted by $\mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$. Its probability density function is given by

$$f_{\boldsymbol{X}}(\mathbf{x}) = (2\pi)^{-\frac{d}{2}} |\boldsymbol{\Sigma}|^{-\frac{1}{2}} \exp\left(-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu})^T \boldsymbol{\Sigma}^{-1}(\mathbf{x} - \boldsymbol{\mu})\right) ,$$

where $\mathbf{x} \in \mathbb{R}^d$.

Q-function is defined as

$$Q(x) = \frac{1}{\sqrt{2\pi}} \int_x^{\infty} e^{-z^2/2} dz$$

For large x, we have $Q(x) \approx \frac{1}{\sqrt{2\pi}x} e^{-x^2/2}$

Hilbert Transform of a signal $g(t)$ is given by

$$\hat{g}(t) = \frac{1}{\pi} \int_{-\infty}^{\infty} \frac{g(\tau)}{t - \tau} d\tau$$

1. a) State whether each of the following statements is true or false, and justify your answer:

i) A random process that is not third-order stationary can be second-order stationary. [2]

ii) Conditioning on an event Y decreases the probability of event Z. [2]

iii) Conditioning on an event Y increases the entropy of a random variable X. [2]

iv) A wide-sense stationary Gaussian process is always white. [2]

v) The power spectral density (PSD) of a first-order stationary process is given by the Fourier transform of its autocorrelation function. [2]

vi) Taking the integral of a wide sense stationary Gaussian process results in a strict sense stationary process. [2]

b) Consider the input signal $m(t) = C \cdot \cos(2\pi f_c t + \Theta)$, where $\Theta = \tan^{-1}(-B/A)$ for some constants A, B, and C. This signal is sampled and the samples are quantized using a uniform quantizer with 2^{n+1} levels. The dynamic range of the quantizer is set to 120% of the quantized signal's peak-to-peak amplitude.

i) What is the minimum required sampling rate? [1]

ii) What is the signal power? [2]

iii) If there is a DC offset of D added to the signal before quantization, what is the quantization noise power in terms of A, B, D, and n? [2]

iv) What is the signal-to-noise ratio (SNR) at the output of the quantizer? [2]

v) What should n be to achieve an SNR of 67.78 dB? [2]

c) Let $\boldsymbol{X} \in \mathbb{R}^n$ be a n-variate Gaussian random variable with mean vector $\boldsymbol{\mu}$ and covariance matrix $\boldsymbol{\Sigma}$, i.e., $\boldsymbol{X} \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$. Let

$$\boldsymbol{Y} = \boldsymbol{A}\boldsymbol{X} - \boldsymbol{b},$$

where $\boldsymbol{Y} \in \mathbb{R}^m$, $\boldsymbol{A} \in \mathbb{R}^{m \times n}$, and $\boldsymbol{b} \in \mathbb{R}^m$. Write down the probability density function of $\boldsymbol{Y}$. [3]

d) Consider a memoryless binary symmetric channel (BSC) with cross-over probability p; that is, if $X \in \{0,1\}$ and $Y \in \{0,1\}$ denote the input and output binary variables for this channel, respectively, we have $P(Y=1|X=0) = P(Y=0|X=1) = p$.

i) Find the channel capacity for $p = 0.3$ and $p = 0.8$. [3]

ii) Can the capacity of the channel be higher when the error probability p is higher? Explain your answer. [3]

e) We consider amplitude modulation of a baseband signal $m(t)$. The signal is first modulated with a waveform $\cos(2\pi f_1 t)$. The modulated waveform is corrupted by additive zero-mean white Gaussian noise $n(t)$ with power spectral density

$$S_n(f) = \begin{cases} 2, & |f| \leq 100 \\ 1, & 100 \leq |f| \leq 200 \\ 0, & \text{otherwise.} \end{cases}$$

Message $m(t)$ is a zero-mean wide sense stationary process with power spectral density given by

$$S_m(f) = \begin{cases} \frac{1}{W}\left(1 - \frac{|f|}{W}\right), & |f| \leq W \\ 0, & \text{otherwise.} \end{cases}$$

The receiver first applies a bandpass filter $H_b(f)$ with the following characteristics

$$H_b(f) = \begin{cases} 1, & f_2 - W \leq |f| \leq f_2 + W \\ 0, & \text{otherwise.} \end{cases}$$

Then, it multiplies the output of the filter with a waveform of $2\cos(2\pi f_2 t + \phi)$. Finally, it applies the following lowpass filter $H_l(f)$ to the signal:

$$H_l(f) = \begin{cases} 1, & |f| \leq W \\ 0, & \text{otherwise.} \end{cases}$$

Let $y(t)$ denote the output of the filter.

i) What is the power of noise $n(t)$? [2]

ii) Find the SNR of $y(t)$ if $f_1 = f_2 = \phi = 0$ and $W = 200$. [2]

iii) Find the SNR of $y(t)$ if $f_1 = f_2 = 50$, $W = 5$ and $\phi = 0$. [3]

iv) Find the SNR of $y(t)$ if $f_1 = f_2 = 150$, $W = 20$ and $\phi = \pi/2$. [3]

2. a) Consider a ternary communication system that transmits a message i randomly drawn from the message set $\{1,2,3\}$ using the signal

$$s_i(t) = A_i q(t), \quad 0 \le t \le T,\ i \in \{1,2,3\},$$

where $A_1 = -A$, $A_2 = 0$, $A_3 = A$, and $q(t)$ is a rectangular pulse:

$$q(t) = \begin{cases} 1, & 0 \le t \le T \\ 0, & \text{otherwise.} \end{cases}$$

Signal is transmitted through an additive white Gaussian noise (AWGN) channel:

$$r(t) = s_i(t) + n(t),$$

where $n(t)$ is the zero-mean noise with power spectral density of $N_0/2$.

The receiver uses a matched filter followed by threshold detection. A decision of "1" is made if the matched filter output is below $-\lambda$ for some $\lambda \ge 0$, decision of "3" is made if it is above $+\lambda$, and a decision of "2" is made otherwise.

i) Find the distribution of the matched filter output at time T for each transmitted signal $s_i(t)$, for $i = 1,2,3$. [3]

ii) Denote by $P_{e,i}$ the probability of error when message "i" is transmitted for $i = 1,2,3$. Derive the expression of $P_{e,i}$ for each i and express your answers in terms of the Q-function. [6]

iii) Denote by p_i the probability of transmitting message i. Let $p_1 = p_3 = 1/4$ and $p_2 = 1/2$. Assume $A = 1$, $N_0 = 1$, and $T = 1$. Derive the optimal threshold λ that minimizes the overall probability of error in the system, and compute the total probability of error under this optimal threshold (express your answer in terms of the Q-function). [4]

iv) Suppose that you want to construct a binary communication system by selecting two signals out of $\{s_1(t), s_2(t), s_3(t)\}$. That is, you want to transmit a message j randomly drawn from the set $\{1,2\}$ using two of the three signals. Which two signals will you choose in order to minimize the error probability? Assuming that the two messages are

equiprobable, derive the optimal threshold detection rule for this system. [3]

b) Consider a digital communication system that employs a 4-QAM modulation scheme to transmit data over an AWGN (Additive White Gaussian Noise) channel with a noise power spectral density of $N_0/2$. The transmitted 4-QAM signal has a minimum distance between constellation points of $2A$, where A is the amplitude of the in-phase and quadrature components. Assume the system uses coherent demodulation at the receiver.

i) Sketch the 4-QAM constellation diagram, label the coordinates of each symbol, and compute the energy of each symbol. [2]

ii) Derive the expression for the symbol error probability P_s in terms of the Q-function. [4]

iii) Assume the receiver experiences a phase synchronization error of ϕ due to imperfect carrier recovery. Assume the transmitted signal is $s(t) = A\sin(2\pi f_c t) + A\cos(2\pi f_c t)$ for one symbol. Derive the effect of this phase error on the decision variables (I and Q components) at the receiver after coherent demodulation. [4]

iv) Derive the symbol error probability for the system described in part (iii), and discuss how the synchronization error affects the symbol error probability. [4]

3. a) Assume that a memoryless source S outputs 5 symbols $\{a,b,c,d,e\}$ with the following probabilities $\Pr(S=a)=0.1$, $\Pr(S=b)=0.15$, $\Pr(S=c)=0.3$, $\Pr(S=d)=0.15$, and $\Pr(S=e)=0.3$.

i) Design a prefix-free code for source S with the minimum possible average codeword length. What is the average length of your code? Is there such a unique code? [5]

ii) What is the entropy of this source? (Please use the approximations $\log_2 3 = 1.58$ and $\log_2 5 = 2.32$ in your calculations). [3]

iii) How can you improve the compression efficiency of this source (i.e., average bits per source symbol)? At most what percentage can you reduce the average code length if you use a more efficient compression technique? [3]

iv) This same code you designed is used for a different source T with the following probabilities: $\Pr(T=a)=0.3$, $\Pr(T=b)=0.25$, $\Pr(T=c)=0.2$, $\Pr(T=d)=0.15$, and $\Pr(T=e)=0.1$. What will be the average codeword length for source T? [3]

v) If each symbol is chosen from either source S or source T with equal likelihood, would you still use the above code, or design a different code? What is the minimum average codeword length one can achieve in this case? [3]

b) Two players, Ayla and Leo, have equal chances of winning each game they play against each other. They play until either Ayla wins a total of 2 games, or Leo wins a total of 3 games. Let X be the random variable denoting the outcome, and Y the number of games played. For example, possible (X,Y) pairs can be $(ALA,3)$, $(ALLA,4)$, or $(LLAL,4)$.

i) Find the entropy of random variable X, denoted by $H(X)$. [4]

ii) Find the entropy of random variable Y, denoted by $H(Y)$. [3]

iii) Find the conditional entropy $H(Y|X)$. [2]

iv) Find the conditional entropy $H(X|Y)$. [4]